

CHE 201 Fall 2019 Quiz 1

Angelo Giles

TOTAL POINTS

45 / 100

QUESTION 1

1 Question 1 **40 / 40**

✓ - **0 pts** Correct

- **10 pts** Patm is wrong
- **5 pts** unit conversion
- **15 pts** Pgage is wrong or not calculated
- **15 pts** P absolute is wrong or not calculated
- **40 pts** Not done or incorrect

QUESTION 2

Question 2 60 pts

2.1 a **5 / 30**

- **0 pts** Correct
- **5 pts** system not identified
- **5 pts** engineering model is missing
- ✓ - **10 pts** PV diagram is wrong
- ✓ - **15 pts** Work calculation is wrong
- **5 pts** unit is not converted
- **30 pts** not done

2.2 b **0 / 30**

- **0 pts** Correct
- ✓ - **20 pts** incorrect or no calculation
- ✓ - **10 pts** PV diagram is wrong
- **0 pts** Click here to replace this description.

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QUIZ 1: 2 problems (10 pts)

Closed book/Closed notes: Show all your work for full credit

1. (6 pts) The average atmospheric pressure on Earth is approximated as a function of altitude by the relation $P_{\text{atm}} = 101.325(1 - 0.02256z)^{5.256}$, where P_{atm} is atmospheric pressure in kPa and z is altitude in km with $z=0$ at sea level. A monometer is used to measure the pressure of a gas in a tank in Denver ($z=1610\text{m}$). The fluid has a density of 850kg/m^3 and the monometer column height is 200cm . Determine the absolute and gage pressure of gas within the tank.

$$P_{\text{atm}} = 101.325(1 - 0.02256z)^{5.256}$$

closed system

$$P_{\text{abs}} = P_{\text{atm}} + P_{\text{gage}}?$$

$$\rho = 850 \text{ kg/m}^3$$

$$g = 9.81 \text{ m/s}^2$$

$$L = 200 \text{ cm}$$

$$P_{\text{gage}} = \rho gh$$

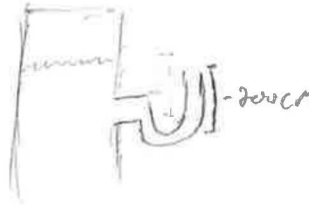
$$z = 1610 \text{ m} = 1.61 \text{ km}$$

$$\begin{aligned} P_{\text{atm}} &= 101.325(1 - 0.02256(1.61))^{5.256} \\ &= 101.325(0.8636784)^{5.256} \\ &= 83.41 \text{ kPa} \end{aligned}$$

$$\begin{aligned} P_{\text{gage}} &= (850 \text{ kg/m}^3)(9.81 \text{ m/s}^2)(2 \text{ m}) \\ &= 16,677 \text{ Pa} \\ &= 16.68 \text{ kPa} \end{aligned}$$

$$P_{\text{abs}} = 83.41 \text{ kPa} + 16.68 \text{ kPa}$$

$$P = 100.1 \text{ kPa}$$



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2. (4 pts) A gas contained within a piston-cylinder assembly undergoes an expansion process for which the relationship between pressure and volume is given by

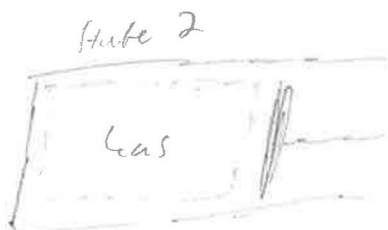
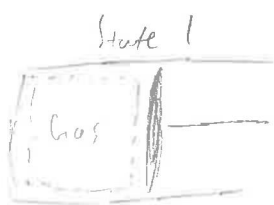
$$PV^n = \text{constant}$$

The initial pressure is 2 bar, $V_1 = 2.5 \text{ m}^3$ to $V_2 = 4 \text{ m}^3$. Determine the work for the process in kJ, if

a. $n = 0$ and

b. $n = 1.5$

c. Sketch both processes on a P-V diagram



closed system

$$P_1 = 2 \text{ bar}$$

$$V_1 = 2.5 \text{ m}^3$$

$$V_2 = 4 \text{ m}^3$$

$$P = \frac{\text{constant}}{V^n}$$

$$P_1 V_1^n = P_2 V_2^n$$

$$(2 \text{ bar})(2.5 \text{ m}^3)^0 = P_2 (4 \text{ m}^3)^0$$

$$2 \text{ bar} = P_2 \quad \text{constant pressure}$$

$$(2 \text{ bar})(2.5 \text{ m}^3)^{3/2} = P_2 (4 \text{ m}^3)^{3/2}$$

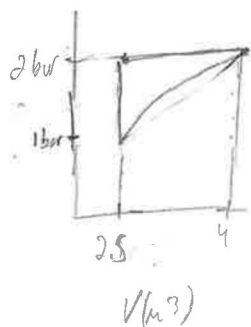
$$\frac{(2 \text{ bar})(3.95 \text{ m}^3)}{8 \text{ m}^3} = \frac{P_2 (8 \text{ m}^3)}{8 \text{ m}^3}$$

$$0.9875 \text{ bar} = P_2$$

$$PV^n = \text{constant}$$

$$W = \int_{V_1}^{V_2} P dV$$

1)



W =

b)